

Daniel Lane

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Summary

PhD in Mechanical Engineering specializing in autonomous systems, perception, and real-time sensor fusion. Research focused on LiDAR-camera fusion for maritime autonomy, embedded perception pipelines, and data-driven modeling of nonlinear dynamical systems. Background includes enterprise-scale system integration and technical leadership, enabling translation between research prototypes and real-world autonomous platforms.

Education

Embry-Riddle Aeronautical University – PhD in Mechanical Engineering Dec 2025
Dissertation: "A Study in Object Detection and Classification Performance by Sensing Modality for Autonomous Surface Vessels"

Embry-Riddle Aeronautical University – MS in Mechanical Engineering May 2021

Stetson University – BS in Physics May 2009

Core Engineering Competencies

Software Development & Project Management: Python, C, C++, HTML/CSS/JavaScript, ROS/ROS2, Git

Machine Learning & Computer Vision: TensorFlow, PyTorch, OpenCV, YOLO, neural network architectures, transfer learning, reinforcement learning

Software & Tools: MATLAB/Simulink, NumPy/Pandas, Gazebo/RViz, Fusion 360, SolidWorks, 3D printing

Robotics & Perception: Sensor fusion, object detection/classification, camera/LiDAR calibration and time synchronization, GNSS/INS, embedded real-time systems, microcontrollers, navigation and path planning

Modeling & Control: Nonlinear dynamic systems, system identification, Kalman filters, information-theoretic methods, PID/LQR control

Engineering Projects & Research

LiDAR-Camera Fusion for Maritime Autonomy

ASV Perception Research, Robotics & Autonomous Systems Lab (RASL), ERAU | NEEC/ONR Grant 2024 – 2025

- Developed lightweight late-fusion method for multi-modal object detection on embedded platforms, agnostic to sensor modality, detection method, and application domain
- Developed custom GStreamer pipeline for sub-millisecond timestamp embedding in video streams, enabling frame-accurate LiDAR-camera synchronization on embedded platforms.

Maritime RobotX Challenge & RoboBoat Challenge – Lead Perception Engineer

Sr. Software Team Member 2024 – 2025

- Developed real-time LiDAR-HDR fusion pipelines for object detection and classification, enabling autonomous mission planning on our ASV for 2024 and 2025 international competitions
- Designed ground control interfaces and sensor pipelines tailored to mission constraints and real-time performance requirements
- Collaborated on testing and validation of perception algorithms in challenging real-world marine environments

Consultation for Yamaha - Future Boat Concept

Co-Researcher 2024

- Collaborated under NDA with Yamaha Marine Division to develop high-level design concepts for future vessels based on feasibility analysis of emerging marine technologies

Data-Driven Modeling of Traffic Systems

Researcher, Complex Dynamical Systems Lab (CDSL), ERAU | NSF Grant 2022 – 2023

- Developed framework for recovering nonlinear system dynamics by detecting model structure through data-driven and information-theoretic methods
- Co-authored peer-reviewed journal publication and presented findings at international MECC 2023 conference
- Supervised undergraduate team on hardware integration and software testing, supporting experimental validation of modeling framework

Publications & Presentations

- D. Lane, "A Study in Object Detection and Classification Performance by Sensing Modality for Autonomous Surface Vessels" (2025). Doctoral Dissertations and Master's Theses. 937. <https://commons.erau.edu/edt/937>.
- D. Lane and S. Roy, "Validating a data-driven framework for vehicular traffic modeling," *Journal of Physics: Complexity*, vol. 5, no. 2, p. 025008, May 2024. doi: 10.1088/2632-072X/ad3ed6.
- D. Lane and S. Roy, "Using information theory to detect model structure with application in vehicular traffic systems," *IFAC-PapersOnLine*, vol. 56, no. 3, pp. 367–372, 2023. 3rd Modeling, Estimation and Control Conference (MECC 2023). doi: 10.1016/j.ifacol.2023.12.051.

Professional Experience

Note: Transitioned from IT systems leadership to graduate research in mechanical engineering and robotics in 2017.

Imaging Systems Specialist, Embry-Riddle Aeronautical University – Daytona Beach, FL 2017 – 2020

- Coordinated software development efforts between internal functional experts and external software development teams to automate document workflows with OCR
- Integrated admissions and registrar systems to streamline student document processing, reducing manual work from hours to minutes

Technology Project Manager, FDOT - Turnpike – Ocoee, FL 2014 – 2017

- Coordinated development and deployment of enterprise-level systems including public-facing web platforms
- Coordinated stakeholder requirements and software vendor efforts to deliver robust IT infrastructure
- Supported SharePoint integration and streamlined internal business process documentation

Senior Technical Roles 2009 – 2014

Apple Inc., Stetson University

- Managed enterprise IT infrastructure, technical troubleshooting, and multimedia systems across multiple organizations
- Built foundational expertise in systems integration, documentation, and automation workflows directly applicable to current robotics R&D and laboratory setup

Additional Highlights

- U.S. Citizen (ITAR eligible)
- Experience leading research teams, integrating autonomous systems, and delivering technical presentations
- Strong foundation in technical documentation and interdisciplinary communication
- Comfortable bridging technical and operational perspectives across academia and industry